A self-similar construction for the Lorentz interpolation condition

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Abstract

An explicit continuous real-valued nonodd function on $[-1,1]$ is constructed whose best uniform polynomial approximant of every degree vanishes at the origin. The function is a geometrically convergent Chebyshev series with a degree-seven seed. The proof has two parts: a finite algebraic certificate for degrees zero through six, and an analytic estimate showing that this certificate holds on the whole interval. Chebyshev composition then propagates the result to every degree. The finite certificate is verified by a self-contained Python program using exact rational arithmetic.

1 The construction

For $f \in C[-1,1]$, let $P_n f$ denote its unique best uniform approximant from the real polynomials of degree at most n, and put $E_n(f) = \|f - P_n f\|_\infty$. Let T_k be the Chebyshev polynomial satisfying $T_k(\cos\theta) = \cos(k\theta)$.

Lorentz's origin interpolation implication, as formulated in [1], asks whether $(P_n f)(0) = 0$ for every n forces f to be odd. The construction below addresses exactly this condition.

Set $z = 2\cos(\pi/36)$ and define the exact real algebraic number

$$\beta = \frac{1}{107}\big(-60 + 82z + 1086z^2 + 1576z^3 - 2932z^4 - 3433z^5 + 2002z^6 + 2243z^7 - 510z^8 - 587z^9 + 44z^{10} + 54z^{11}\big). \tag{1}$$

The certificate below gives $0.419 < \beta < 0.420$; its decimal value is $0.4192188126781505\ldots$. Define

$$h = T_4 - 1 - 2T_5 + \beta T_7, \qquad \varepsilon = 10^{-10}, \qquad q_\varepsilon = T_3 + \varepsilon h, \tag{2}$$

and

$$f_\varepsilon(x) = \sum_{j=0}^{\infty} \left(-\frac{1}{3}\right)^j q_\varepsilon(T_{3^j}(x)). \tag{3}$$

The series converges uniformly, since $T_k([-1,1]) = [-1,1]$ and $\sum_j 3^{-j} < \infty$. Thus f_ε is real and continuous.

Theorem 1.1. *The function in* (3) *satisfies*

$$(P_n f_\varepsilon)(0) = 0 \qquad (n = 0, 1, 2, \ldots),$$

but is not odd. More precisely, with $a = 1/\sqrt{2}$,

$$f_\varepsilon(a) + f_\varepsilon(-a) = -3\varepsilon.$$

The exact value of β, rather than a rounded decimal, is part of the definition. The small value of ε is chosen to make the analytic estimates particularly conservative.

2 Why seven conditions suffice

Lemma 2.1 (Chebyshev composition)**.** *For $g \in C[-1,1]$, an integer $M \geq 1$, and $n \geq 0$,*

$$P_n(g \circ T_M) = (P_{\lfloor n/M \rfloor} g) \circ T_M.$$

Proof. Write $x = \cos\theta$. Given a polynomial competitor $p \in \mathcal{P}_n$, average $p(\cos\theta)$ over the shifts $\theta \mapsto \theta + 2\pi j/M$, $0 \leq j < M$. The target $g(\cos M\theta)$ is unchanged by these shifts. The average has no larger uniform error, and only frequencies divisible by M remain. It therefore has the form $r(\cos M\theta)$ with $r \in \mathcal{P}_{\lfloor n/M \rfloor}$. Since T_M maps $[-1,1]$ onto itself, its error is exactly $\|g - r\|_\infty$. The two minima are equal, and uniqueness gives the asserted identity. □

The series (3) obeys

$$f_\varepsilon = q_\varepsilon - \frac{1}{3} f_\varepsilon \circ T_3. \tag{4}$$

Polynomial translation of best approximation, homogeneity, and Lemma 2.1 imply, for every $n \geq 7$,

$$P_n f_\varepsilon = q_\varepsilon - \frac{1}{3}(P_{\lfloor n/3 \rfloor} f_\varepsilon) \circ T_3. \tag{5}$$

Both $q_\varepsilon(0)$ and $T_3(0)$ are zero. Hence

$$(P_n f_\varepsilon)(0) = -\frac{1}{3}(P_{\lfloor n/3 \rfloor} f_\varepsilon)(0), \qquad n \geq 7. \tag{6}$$

It remains to establish the seven cases $0 \leq n \leq 6$.

3 The odd function at the center of the construction

Define

$$G(x) = \sum_{j \geq 0} (-1/3)^j T_{3^j}(x), \qquad F = G \circ T_3, \qquad H = \sum_{j \geq 0} (-1/3)^j h \circ T_{3^j}.$$

Thus $f_\varepsilon = F + \varepsilon H$. All these series converge uniformly. The functions G and F are odd, whereas the even part of H is nonzero.

Put $a = 1/\sqrt{2}$ and

$$g(x) = x - \tfrac{1}{3} T_3(x) = 2x - \tfrac{4}{3} x^3.$$

Pairing consecutive terms gives

$$G = \sum_{j \geq 0} 9^{-j} g \circ T_{9^j}. \tag{7}$$

The polynomial g has its unique maximum at a, unique minimum at $-a$, and $T_9(\pm a) = \pm a$. Consequently

$$B := \|G\|_\infty = \frac{g(a)}{1 - 1/9} = \frac{3}{2\sqrt{2}},$$

and the only points at which $|G| = B$ are $\pm a$.

Since $P_0 G = 0$ and $G = x - \frac{1}{3} G \circ T_3$, Lemma 2.1 gives $P_1 G = P_2 G = x$. Therefore

$$P_n F = \begin{cases} 0, & 0 \leq n \leq 2, \\ T_3, & 3 \leq n \leq 6, \end{cases} \qquad E_n(F) = \begin{cases} B, & 0 \leq n \leq 2, \\ B/3, & 3 \leq n \leq 6. \end{cases} \tag{8}$$

The extremal sets of these two residuals are respectively the zeros of T_6 and T_{18}. They are finite, and their exact locations make the calculation in the next section possible.

4 The exact finite certificate

This section verifies the seven finite conditions used in the construction. All coefficients and all nodes appearing below are algebraic. The verification uses rational arithmetic in a number field, together with rational interval bounds for strict inequalities; it does not use a sampled approximation to the infinite series.

Put $a = 1/\sqrt{2}$, and retain the definitions

$$h = T_4 - 1 - 2T_5 + \beta T_7, \qquad H(x) = \sum_{j=0}^{\infty} (-1/3)^j h(T_{3^j}(x)).$$

Here β is the explicit algebraic number specified in the construction. Define the increasing fine grid by

$$X_j = -\cos\frac{(2j-1)\pi}{36}, \qquad 1 \le j \le 18, \qquad \mathcal{F} = \{X_1, \dots, X_{18}\}.$$

The coarse grid, also in increasing order, is

$$(C_1, C_2, C_3, C_4, C_5, C_6) = (X_2, X_5, X_8, X_{11}, X_{14}, X_{17}), \qquad \mathcal{C} = \{C_1, \dots, C_6\}.$$

Thus $\mathcal{F}$ is the zero set of T_{18} and $\mathcal{C}$ is the zero set of T_6. Let σ_j repeat the four signs $(+,-,-,+)$ along $\mathcal{F}$, and let τ_j repeat $(-,+,+,-)$ along $\mathcal{C}$. Equivalently,

$$\sigma_j = -\operatorname{sgn} T_9(X_j), \qquad \tau_j = \operatorname{sgn} T_3(C_j).$$

These are precisely the signs of the unperturbed residuals on the two grids.

4.1 Exact values of the perturbation on the grids

Since $T_3(a) = -a$ and $T_3(-a) = a$, the functional equation $H = h - H \circ T_3/3$ gives

$$H(a) = -\frac{3}{2} + \frac{3}{2}(2+\beta)a, \qquad H(-a) = -\frac{3}{2} - \frac{3}{2}(2+\beta)a.$$

For $x \in \mathcal{F}$ we have $T_9(x) \in \{-a, a\}$. Iterating the functional equation twice therefore gives

$$\begin{aligned} H(x) &= h(x) - \frac{1}{3}h(T_3(x)) + \frac{1}{9}H(T_9(x)) \\ &= h(x) - \frac{1}{3}h(T_3(x)) - \frac{1}{6} + \frac{2+\beta}{6}T_9(x). \end{aligned}$$

On this grid, $T_{21}(x) = -T_{15}(x)$, by the cosine addition formula and $T_{18}(x) = 0$. Consequently

$$H(X_j) = R_\beta(X_j), \qquad 1 \le j \le 18, \tag{9}$$

where the explicitly known polynomial is

$$\begin{aligned} R_\beta(x) = T_4(x) - \frac{1}{3}T_{12}(x) - \frac{5}{6} - 2T_5(x) + \beta T_7(x) \\ + \frac{2+\beta}{3}T_{15}(x) + \frac{2+\beta}{6}T_9(x). \end{aligned} \tag{10}$$

The same formula supplies the values at every coarse node, since $\mathcal{C} \subset \mathcal{F}$.

4.2 Exact definitions of the seven polynomial corrections

We define polynomials $v_n \in \mathcal{P}_n$ and positive real numbers λ_n, for $0 \le n \le 6$. First set

$$v_0 = 0, \qquad \lambda_0 = \frac{\sqrt{6}}{4}(2+\beta),$$
$$v_1 = v_2 = -(2+\beta)x, \qquad \lambda_1 = \lambda_2 = \frac{2+\beta}{2\sqrt{2}}. \tag{11}$$

Write

$$v_3(x) = A_1 x + A_2 x^2 + A_3 x^3.$$

Define A_1, A_2, A_3, λ_3 by the four linear equations

$$v_3(X_j) + \sigma_j \lambda_3 = R_\beta(X_j), \qquad j \in \{1, 3, 8, 10\}. \tag{12}$$

The matrix of this system is nonsingular. The choice of β also makes the same equality hold at $j = 16$; this fifth equality is included in the exact certificate below.

Next write

$$v_4(x) = T_4(x) - 1 + B_1 T_1(x) + B_3 T_3(x),$$

and define B_1, B_3, λ_4 by

$$v_4(X_j) + \sigma_j \lambda_4 = R_\beta(X_j), \qquad j \in \{10, 16, 18\}. \tag{13}$$

Finally set

$$v_5(x) = v_6(x) = T_4(x) - 1 + D_1 T_1(x) + D_3 T_3(x) + D_5 T_5(x), \qquad \lambda_6 = \lambda_5,$$

where D_1, D_3, D_5, λ_5 solve

$$v_5(X_j) + \sigma_j \lambda_5 = R_\beta(X_j), \qquad j \in \{10, 13, 15, 16\}. \tag{14}$$

The matrices in (13) and (14) are also nonsingular. These small linear systems, together with the explicit value of β, are exact definitions of all the coefficients. In particular, they avoid the use of decimal coefficients in the construction.

By their displayed forms, the corrections satisfy

$$\deg v_n \le n, \qquad v_n(0) = 0 \qquad (0 \le n \le 6). \tag{15}$$

Here $T_4(0) = 1$ and all the remaining Chebyshev polynomials appearing in v_4, v_5, v_6 are odd.

4.3 Inequalities and alternating equality sets

For $n = 0, 1, 2$, define the coarse slacks

$$s_{n,j} = \lambda_n - \tau_j\big(H(C_j) - v_n(C_j)\big), \qquad 1 \le j \le 6.$$

For $3 \le n \le 6$, define the fine slacks

$$s_{n,j} = \lambda_n - \sigma_j\big(H(X_j) - v_n(X_j)\big), \qquad 1 \le j \le 18.$$

The following table gives the complete sets J_n of zero slacks. Every slack outside the listed set is strictly greater than the rational lower bound in the last column.

n	grid	$J_n = \{j : s_{n,j} = 0\}$	lower bound if $j \notin J_n$
0	$\mathcal{C}$	$3, 4$	$127/50$
1	$\mathcal{C}$	$1, 3, 4, 6$	$363/100$
2	$\mathcal{C}$	$1, 3, 4, 6$	$363/100$
3	$\mathcal{F}$	$1, 3, 8, 10, 16$	$9/100$
4	$\mathcal{F}$	$1, 3, 8, 9, 10, 16, 18$	$7/10$
5	$\mathcal{F}$	$3, 4, 6, 9, 10, 13, 15, 16$	$1/5$
6	$\mathcal{F}$	$3, 4, 6, 9, 10, 13, 15, 16$	$1/5$

In particular, the equality at $j = 8$ in the row $n = 4$ is an exact algebraic identity, not an equality inferred from a small numerical residual.

For $n = 0, 2, 3, 6$, use every index in J_n as the alternating equality set. For $n = 1$, use $1, 3, 4$; for $n = 4$, use $1, 3, 9, 10, 16, 18$; and for $n = 5$, use $3, 4, 6, 9, 10, 13, 15$. Each selected set contains exactly $n + 2$ nodes, in increasing order, and its indicated signs alternate. We have consequently verified, at the finite grids,

$$\begin{cases} \tau_j\big(H(C_j) - v_n(C_j)\big) \le \lambda_n, & 0 \le n \le 2, \\ \sigma_j\big(H(X_j) - v_n(X_j)\big) \le \lambda_n, & 3 \le n \le 6, \end{cases} \tag{16}$$

with $n + 2$ alternating equalities in each case. These are signed inequalities for the perturbation at the extrema of the unperturbed residual. The separate persistence argument converts them into uniform error bounds on the entire interval for the perturbed residuals.

4.4 How the finite assertions are certified

All the preceding assertions are verified by the accompanying `verify_lorentz.py` script. It uses only Python's standard library and exact rational arithmetic. For completeness, its number field and verification procedure are specified here.

Set $z = 2\cos(\pi/36)$. Arithmetic is performed modulo

$$m(z) = z^{12} - 12z^{10} + 54z^8 - 112z^6 + 105z^4 - 36z^2 + 1. \tag{17}$$

Indeed,

$$m(z) = (z^6 - 6z^4 + 9z^2 - 2)^2 - 3.$$

Every $T_k(X_j)$ is obtained from

$$T_k(X_j) = (-1)^k \cos\frac{k(2j-1)\pi}{36},$$

using the integer recurrence for $2\cos(k\pi/36)$ and reduction modulo m. Gaussian elimination and polynomial Euclidean inversion solve the three defining linear systems exactly. The code also verifies that the resulting β equals its explicit polynomial expression in z.

The real embedding is fixed by $1.98 < z < 2$. There is exactly one root of m in this interval, as checked by an exact Sturm count. One can also see uniqueness directly: if $S_6(z) = z^6 - 6z^4 + 9z^2 - 2$, then

$$S_6'(z) = 6z(z^2 - 1)(z^2 - 3) > 0 \quad (1.98 \le z \le 2), \qquad S_6(1.98) > 0,$$

so m is strictly increasing there. The endpoint signs are checked using rational arithmetic. Three hundred rational bisection steps give an isolating interval. Evaluating each reduced polynomial on this interval by rational interval arithmetic certifies every strict inequality in the table. Every reported zero is first checked as the zero polynomial in the quotient field. Thus no tolerance is used to classify an equality as exact.

The same calculation gives

$$\frac{419}{1000} < \beta < \frac{420}{1000}, \qquad 0 < \lambda_n < 2 \quad (0 \le n \le 6).$$

If $\|v\|_{\mathrm{Ch}}$ denotes the sum of the absolute values of the Chebyshev coefficients of a polynomial v, the certified bounds are

n	0	1	2	3	4	5	6
$\|v_n\|_{\mathrm{Ch}}$	0	$< 5/2$	$< 5/2$	$< 9/4$	$< 13/4$	$< 11/2$	$< 11/2.$

In particular, $\|v_n\|_\infty < 6$. Since $\|T_k'\|_\infty = k^2$ and $\deg v_n \le 6$, these coefficient bounds also give $\|v_n'\|_\infty < 216$. Both bounds are independently asserted in the exact certificate. They supply the uniform constants needed in the persistence argument.

5 From the finite certificate to uniform minimax approximation

We use precisely the following properties of the finite certificate. For each $0 \le n \le 6$, it supplies $v_n \in \mathcal{P}_n$ and $\lambda_n \in \mathbb{R}$. Every v_n vanishes at zero, and the sum of the absolute values of its Chebyshev coefficients is less than 6. Set

$$p_n^0 = \begin{cases} 0, & 0 \le n \le 2, \\ T_3, & 3 \le n \le 6, \end{cases} \qquad b_n = F - p_n^0.$$

The associated finite sets are

$$\mathcal{X}_n = \begin{cases} \{x : T_6(x) = 0\}, & 0 \le n \le 2, \\ \{x : T_{18}(x) = 0\}, & 3 \le n \le 6. \end{cases}$$

For $z \in \mathcal{X}_n$, put $\sigma_z = \operatorname{sgn} b_n(z)$. The certificate asserts

$$\sigma_z\big(H(z) - v_n(z)\big) \le \lambda_n \qquad (z \in \mathcal{X}_n), \tag{18}$$

and gives $n+2$ increasing points of $\mathcal{X}_n$ at which equality holds and the signs σ_z alternate. These are finite assertions about exact algebraic numbers. The argument below proves that they also certify the minimax problem on the whole interval, for the explicit choice

$$\varepsilon = 10^{-10}.$$

5.1 A uniform linear bound for the loss from a maximum

The identities $T_3(a) = -a$ and $T_9(a) = a$ give

$$E := G(a) = \frac{g(a)}{1 - 1/9} = \frac{3a}{2}.$$

The following factorization holds throughout $[-1, 1]$:

$$g(a) - g(x) = \frac{4}{3}(x - a)^2(x + \sqrt{2}) \ge C_0(x - a)^2, \qquad C_0 = \frac{4}{3}(\sqrt{2} - 1). \tag{19}$$

Thus g has a unique global maximum at a. Every summand in

$$E - G(x) = \sum_{j=0}^{\infty} 9^{-j}\big(g(a) - g(T_{9^j}(x))\big)$$

is nonnegative. Consequently G also has its unique global maximum at a, and, by oddness, its unique global minimum at $-a$.

We require a stronger estimate than the quadratic bound from the first summand. Set $\delta = \pi/12$ and write $x = \cos(\pi/4 + u)$, initially with $0 < |u| \le \delta$. Choose the largest integer $j \ge 0$ for which $9^j|u| \le \delta$; then

$$\frac{\delta}{9} < 9^j|u| \le \delta.$$

Since $9^j \equiv 1 \pmod 8$,

$$T_{9^j}(x) = \cos(\pi/4 + 9^j u).$$

On $[\pi/6, \pi/3]$, the sine is at least $1/2$. The mean value theorem and (19) therefore imply

$$\begin{aligned} E - G(x) &\ge 9^{-j}\big(g(a) - g(T_{9^j}(x))\big) \\ &\ge \frac{C_0}{4}\, 9^{-j}(9^j u)^2 > \frac{C_0\delta}{36}|u| \ge c_G|x-a|, \qquad c_G = \frac{\pi(\sqrt{2}-1)}{324}. \end{aligned}$$

The same inequality is trivial at $u = 0$. If the angular distance from $\pi/4$ exceeds δ, then

$$|x - a| \ge r_0 := \frac{\sqrt{3} - \sqrt{2}}{2}.$$

The first summand alone now gives $E - G(x) \ge C_0 r_0 |x - a|$. Since $C_0 r_0 > c_G$, the bound holds for every $x \in [-1, 1]$. Reflecting x yields the minimum estimate as well. Hence

$$E - |G(x)| \ge c_G \operatorname{dist}(x, \{-a, a\}) \qquad (-1 \le x \le 1). \tag{20}$$

For a positive integer L, let

$$X_L = \{x : T_{2L}(x) = 0\} = \left\{\cos\frac{(2j-1)\pi}{4L} : 1 \le j \le 2L\right\}.$$

These are exactly the preimages under T_L of $\{-a, a\}$. We shall use

$$\operatorname{dist}(T_L(x), \{-a, a\}) \ge \kappa L \operatorname{dist}(x, X_L), \qquad \kappa = \frac{4(1-a)}{\pi}. \tag{21}$$

To verify it, write $x = \cos\theta$. In the angular variable the nearest zero of T_{2L} is at distance at most $\pi/(4L)$. By the symmetries of the cosine, the needed scalar inequality reduces to

$$|\cos t - a| \ge \kappa |t - \pi/4| \qquad (0 \le t \le \pi/2).$$

On $[0, \pi/4]$ this follows from concavity of the cosine and its chord between the endpoints. On $[\pi/4, \pi/2]$, it follows by integrating $\sin t \ge a > \kappa$. Finally, $|\cos\theta - \cos\phi| \le |\theta - \phi|$ converts angular distance into the distance in (21).

For $0 \le n \le 2$ we have $b_n = G \circ T_3$, whereas for $3 \le n \le 6$,

$$b_n = F - T_3 = -\frac{1}{3} G \circ T_9.$$

Their norms are, respectively, $E_n = E$ and $E_n = E/3$. In particular, $E_n \geq 1/(2\sqrt{2}) > 1/3$. Applying (20) and (21) gives, in both cases,

$$E_n - |b_n(x)| \geq c\,\mathrm{dist}(x, \mathcal{X}_n), \qquad c = 3c_G\kappa = \frac{3-2\sqrt{2}}{27\sqrt{2}} > \frac{1}{250}. \tag{22}$$

The last inequality can be checked without approximation: it is equivalent to $750 > 527\sqrt{2}$, and $750^2 > 2 \cdot 527^2$.

5.2 Pointwise Lipschitz estimates at the required nodes

We do not assume that H is globally Lipschitz. Only its differences from its values at certain fixed points are needed. Pairing the series for H gives

$$H = \sum_{j=0}^{\infty} 9^{-j} Q \circ T_{9^j}, \qquad Q = h - \frac{1}{3} h \circ T_3.$$

Since $T_3(\pm a) = \mp a$ and $T_3'(\pm a) = 3$,

$$Q'(a) = h'(a) - h'(-a) = 2h'_{\mathrm{even}}(a).$$

Here $h_{\mathrm{even}} = T_4 - 1$, and $T_4'(x) = 16x(2x^2 - 1)$, so $Q'(a) = Q'(-a) = 0$.

Let $x = \cos(\pi/4 + u)$ with $0 < |u| \leq \delta = \pi/12$, and choose j as in the preceding subsection. Taylor's theorem at a gives, for $0 \leq k \leq j$,

$$|Q(T_{9^k}(x)) - Q(a)| \leq \frac{1}{2}\|Q''\|_\infty (9^k u)^2.$$

Summing these terms and bounding the remaining tail by $2\|Q\|_\infty$ termwise gives

$$\begin{aligned} |H(x) - H(a)| &\leq \frac{1}{2}\|Q''\|_\infty u^2 \sum_{k=0}^{j} 9^k + 2\|Q\|_\infty \sum_{k=j+1}^{\infty} 9^{-k} \\ &\leq \left(\frac{9\delta}{16}\|Q''\|_\infty + \frac{9}{4\delta}\|Q\|_\infty \right) |u|. \end{aligned}$$

On this angular interval, $|x - a| \geq |u|/2$. Thus

$$|H(x) - H(a)| \leq L_H |x - a|, \qquad L_H = \frac{3\pi}{32}\|Q''\|_\infty + \frac{54}{\pi}\|Q\|_\infty. \tag{23}$$

The identical argument about $3\pi/4$ proves the analogous inequality at $-a$. In particular, (23) applies whenever $|x - a| \leq r_0$, and its reflected version applies whenever $|x + a| \leq r_0$.

For completeness, all needed bounds can be kept deliberately coarse. The Chebyshev coefficient formulas and $\beta < 0.420$ give

$$\|h\|_\infty < 4.42, \qquad \|h'\|_\infty < 16 + 2 \cdot 25 + 0.420 \cdot 49 < 87,$$

and

$$\|h''\|_\infty < 80 + 2 \cdot 200 + 0.420 \cdot 784 < 810.$$

We used $\|T_k'\|_\infty = k^2$ and $\|T_k''\|_\infty = k^2(k^2 - 1)/3$. Since $\|T_3'\|_\infty = 9$ and $\|T_3''\|_\infty = 24$, the chain rule gives

$$\|Q\|_\infty < 5.894, \qquad \|Q''\|_\infty \leq 28\|h''\|_\infty + 8\|h'\|_\infty < 23376.$$

Using $3 < \pi < 22/7$ in (23) consequently yields $L_H < 7000$.

If $L = 3^k$ and $z \in X_L$, iterate the functional equation for H to obtain

$$H(x) = S_k(x) + \left(-\frac{1}{3}\right)^k H(T_L(x)), \qquad S_k(x) = \sum_{j=0}^{k-1} \left(-\frac{1}{3}\right)^j h(T_{3^j}(x)).$$

For $|x - z| \le r_0/L^2$, the bound $\|T_L'\|_\infty = L^2$ permits application of (23) at $T_L(z) = \pm a$. Moreover,

$$\|S_k'\|_\infty \le \|h'\|_\infty \sum_{j=0}^{k-1} 3^j = \frac{L-1}{2}\|h'\|_\infty.$$

It follows that

$$|H(x) - H(z)| \le \left(\frac{L-1}{2}\|h'\|_\infty + LL_H\right)|x - z| \qquad \left(|x - z| \le \frac{r_0}{L^2}\right). \tag{24}$$

For the grids used here, $L = 3$ or 9, and $r_0/L^2 \ge (\sqrt{3} - \sqrt{2})/162 > 10^{-3}$. The coefficient bound on v_n implies $\|v_n\|_\infty < 6$ and $\|v_n'\|_\infty < 6 \cdot 6^2 = 216$. Consequently, with $r_n = H - v_n$, we have the uniform estimates

$$\|r_n\|_\infty < 20, \qquad |r_n(x) - r_n(z)| < 10^5|x - z| \quad (z \in \mathcal{X}_n,\ |x - z| \le 10^{-3}). \tag{25}$$

Indeed, $\|H\|_\infty \le \frac{3}{2}\|h\|_\infty < 6.63$, and the largest Lipschitz bound is less than $4{\cdot}87{+}9{\cdot}7000{+}216 < 10^5$. Since equality occurs in (18), it also follows that $|\lambda_n| < 20$.

5.3 Verification on the entire interval

Fix $0 \le n \le 6$, and consider the candidate

$$p_n = p_n^0 + \varepsilon v_n, \qquad \varepsilon = 10^{-10}.$$

We prove

$$\|f_\varepsilon - p_n\|_\infty \le E_n + \varepsilon\lambda_n. \tag{26}$$

First suppose $\operatorname{dist}(x, \mathcal{X}_n) \le 10^{-3}$, and choose a nearest node z. Write $L = 3$ or 9 according to the grid. Since $|T_L(x) - T_L(z)| \le 81 \cdot 10^{-3} < r_0$, the point $T_L(x)$ lies in $[1/2, \sqrt{3}/2]$ if $T_L(z) = a$, and in the reflected interval if $T_L(z) = -a$. On the positive interval, $g(y) \ge 5/6$. The identity $G = g + \frac{1}{9}G \circ T_9$ therefore gives

$$G(y) \ge \frac{5}{6} - \frac{E}{9} > \frac{2}{3} > \frac{E}{2}.$$

Oddness gives the corresponding negative bound. It follows, for either baseline residual, that

$$\sigma_z b_n(x) > E_n/2.$$

By (25), $\varepsilon|r_n(x)| < 2 \cdot 10^{-9} < E_n/4$. Thus the perturbed residual $b_n(x) + \varepsilon r_n(x)$ has sign σ_z. Put $d = |x - z|$. Equations (22), (18), and (25) now give

$$\begin{aligned} |b_n(x) + \varepsilon r_n(x)| &= \sigma_z b_n(x) + \varepsilon\sigma_z r_n(x) \\ &\le E_n - cd + \varepsilon\lambda_n + \varepsilon 10^5 d \\ &\le E_n + \varepsilon\lambda_n, \end{aligned}$$

because $\varepsilon 10^5 = 10^{-5} < c$.

If instead $\operatorname{dist}(x, \mathcal{X}_n) > 10^{-3}$, then

$$|b_n(x) + \varepsilon r_n(x)| \le E_n - c\,10^{-3} + 20\varepsilon \le E_n - 20\varepsilon \le E_n + \varepsilon\lambda_n.$$

Here we used $40\varepsilon = 4 \cdot 10^{-9} < c\,10^{-3}$ and $\lambda_n > -20$. This proves (26).

At each of the $n+2$ alternating equality nodes supplied by the certificate, the error is exactly

$$b_n(z) + \varepsilon r_n(z) = \sigma_z\big(E_n + \varepsilon\lambda_n\big).$$

The common magnitude is positive, since $E_n > 1/3$ and $\varepsilon|\lambda_n| < 2 \cdot 10^{-9}$. The alternation theorem therefore proves that p_n is the unique best uniform approximant to f_ε from $\mathcal{P}_n$. In particular,

$$(P_n f_\varepsilon)(0) = 0 \qquad (0 \le n \le 6).$$

6 Completion of the construction

The finite certificate and the stability estimate establish

$$P_n f_\varepsilon = P_n F + \varepsilon v_n, \qquad 0 \le n \le 6,$$

where every $v_n(0) = 0$. The baseline approximants in (8) also vanish at zero. Thus all seven base conditions hold. Induction using (6) proves the origin condition for every degree.

For nonoddness, the even part of h is $h_e = T_4 - 1$. At $a = 1/\sqrt{2}$ we have $T_3(a) = -a$, $T_3(-a) = a$, and $h_e(a) = h_e(-a) = -2$. The even part H_e consequently satisfies

$$H_e(a) = \sum_{j \ge 0} (-1/3)^j (-2) = -\frac{3}{2}.$$

Since F is odd,

$$f_\varepsilon(a) + f_\varepsilon(-a) = 2\varepsilon H_e(a) = -3\varepsilon \neq 0.$$

This proves Theorem 1.1.

Reproducing the finite verification

Run `python3 verify_lorentz.py`. The script requires only the Python standard library. It constructs the algebraic coefficients, checks every zero identity and alternation, isolates the real embedding by a Sturm count and rational bisection, and verifies all remaining signs by rational interval arithmetic. It writes `certificate.json`, which records the exact coefficients and all finite-grid slacks. Floating-point conversion is used only to print readable decimal summaries.

The finite computation alone is not the whole-interval claim: the analytic stability argument is essential. Conversely, the stability argument is used only after the exact finite inequalities and alternating equalities have been verified.

References

[1] E. B. Saff and R. S. Varga, *Remarks on some conjectures of G. G. Lorentz*, Journal of Approximation Theory **30** (1980), 29–36. `https://www.math.kent.edu/~varga/pub/paper_126.pdf`.